



Databases and its Applications

Advanced NoSQL and Emerging Database trends

Pooja T S

Computer Applications

Databases and its Applications

Advanced NoSQL and Emerging Database trends

Transactions in MongoDB: ACID, Write Concerns

Pooja T S

Computer Applications



- ▶ MongoDB supports multi-document transactions to ensure data consistency.
- ▶ Transactions allow:
 - Grouping multiple operations
 - Executing them as a single logical unit
 - Either fully committing or fully aborting



Databases and its Applications

ACID Properties in MongoDB

- ▶ MongoDB transactions follow ACID principles:

Atomicity – All operations succeed or none do

Consistency – Database rules remain valid

Isolation – Concurrent transactions do not interfere

Durability – Committed data is permanently stored



When to Use Transactions

- ▶ Transactions are required when:
 - Updates span multiple collections
 - Data consistency across documents is critical
 - Partial updates are unacceptable
- ▶ Example use cases:
 - Order processing
 - Inventory updates



Write Concern

- ▶ Write concern defines how MongoDB confirms write operations.
- ▶ It controls:

Number of nodes acknowledging the write

Whether data is written to disk

Reliability vs performance trade-off



Write Concern Levels

- ▶ Common write concern values:
 - `w:1` – Acknowledged by primary only
 - `w:"majority"` – Acknowledged by majority of nodes
 - `j:true` – Written to journal
- ▶ Stronger write concern increases durability.



Read Concern

- ▶ Read concern controls the visibility of data during reads.
- ▶ It determines:

Whether uncommitted data is visible
Consistency level of read operations



Read Concern Levels

- ▶ Common read concern levels:
 - local – Reads data from local node
 - majority – Reads only committed data
 - snapshot – Consistent view within a transaction
- ▶ Used heavily inside transactions.



Databases and its Applications

Transactions in Mongo Shell

- ▶ In Mongo Shell:
 - A session is started
 - Transaction is initiated within the session
 - Operations are executed
 - Transaction is committed or aborted
- ▶ Used mainly in replica sets.



Databases and its Applications

Transaction Limitations

- ▶ **MongoDB transactions:**
 - Are slower than single-document operations
 - Require replica set or sharded cluster
 - Should be used only when necessary



Databases and its Applications

Summary – MongoDB Transactions

- ▶ Transactions ensure consistency across multiple operations.
- ▶ Key components:
 - ACID guarantees
 - Write concern for durability
 - Read concern for consistency

